

Lab Exercise #4: Utility Tracker

1 Overview

Write a program that tracks a household's utility usage. The program should repeatedly display a menu and allow the user to enter usage data until they choose to exit.

Your program must repeatedly display this menu:

1. Enter electricity usage (kWh)
2. Enter water usage (gallons)
3. Enter gas usage (therms)
4. Display totals
5. Exit

- Use a **while** loop to repeat the menu.
- Use a **switch** statement to handle menu choices.
- Continue running until the user chooses option 5.
- Keep running totals for:
 - Electricity
 - Water
 - Gas
- Validate that all usage inputs are non-negative.
- If the user enters an invalid menu option, display an error message.
- Display Totals Option

When the user selects option 4, display:

- Total electricity usage
- Total water usage
- Total gas usage
- Number of entries made

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #4 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #4 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Lab 4: Mark Schwitzerlett and Jacobi
February 23, 2026

1. Enter electricity usage (kWh)
2. Enter water usage (gallons)
3. Enter gas usage (therms)
4. Display totals
5. Exit

Enter choice: 1
Enter electricity usage: 250

1. Enter electricity usage (kWh)
2. Enter water usage (gallons)
3. Enter gas usage (therms)
4. Display totals
5. Exit

Enter choice: 2
Enter water usage: 800

1. Enter electricity usage (kWh)
2. Enter water usage (gallons)
3. Enter gas usage (therms)
4. Display totals
5. Exit

Enter choice: 4

Totals:
Electricity: 250.00 kWh
Water: 800.00 gallons
Gas: 0.00 therms
Entries made: 2

1. Enter electricity usage (kWh)
2. Enter water usage (gallons)
3. Enter gas usage (therms)
4. Display totals
5. Exit

Enter choice: 5
Program terminated.

...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named Lab#4_FirstName_LastName.c

Due date: The lab is due at the end of your lab period, either Monday, February 23rd or Tuesday, February 24th.

Be sure to save a copy of your .c file.